Problem A. Prime Maze

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	128 megabytes

Cynthia loves maze! After studying prime numbers at school, she decided to design a "prime maze".

Note that a prime number (or a prime) is a natural number greater than 1 that is not a product of two smaller natural numbers.

The maze consists of R rows and C columns, with each cell containing a number, starting from 1, and ends with $R \times C$. To be clear, the cell located at the r-th row and c-th column, denoted as (r, c), contains the number $(r-1) \times C + c$.

Cynthia would construct a seemingly complicated maze by making cells that contain prime numbers inaccessible (obstructed). The following figure shows an example of a 5×8 maze, where the obstacles are colored **black**.

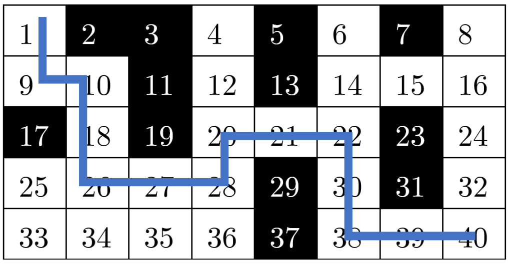

Figure 1: An example prime maze with $R = 5$ and $C = 8$.

We will consider cells in a maze as **adjacent** if they have a common side. That is, cell (r, c) is adjacent to cells $(r, c-1)$, $(r, c+1)$, $(r-1, c)$ and $(r+1, c)$, respectively.

Starting from $(1, 1)$, each time you can take **one** move to an adjacent cell that is not an obstacle. The blue route shown in the example shows a path starting from $(1, 1)$ to $(5, 8)$ in minimum moves, which is 13 in total.

Since Cynthia is an expert, she would like to solve mazes as large as possible. However, she found it hard to draw a maze that is ideal for her. Now, Cynthia wants to obtain the *minimum* move(s) starting from source cell $(1, 1)$ to the destination cell (R, C). Can you help her?

Input

There are multiple test cases. The first line of input contains an integer T ($1 \le T \le 10^6$), indicating the number of test cases. For each test case:

The first and only line contains two integers R and C ($1 \le R \le 2^{31} - 1, 1 \le C \le 20$).

Output

For each test case output one line containing one integer, indicating the minimum move(s) to move from the source cell to the destination cell. Output `-1` if there is no such way.

Example

standard input	standard output
2 2 2 5 8	-1 13

Note

The second test case is exactly the example shown above, and the blue line with minimum moves 13 corresponds to the second output.